

Producing algorithms: combined solutions

Revise 360

Worksheet for the 360° experience: Algorithm construction lab

Name: _____

Class: _____

Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Algorithm construction lab. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Extract requirements from a problem.
- Combine sequence, selection and iteration.
- Refine and test a complete solution.

Starter: before you put the headset on

A club charges 3 per hour for each item. What is the charge for two items hired for 2 and 4 hours?

.....

.....

Key terminology

Requirement

.....

.....

Nested structure

.....

.....

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Read the requirements

Turn right

Key facts in your own words:

Example to trace:

Hire three items
Input hours for each item
Each costs 3 per hour
Output combined charge

Challenge: Which inputs and output are required?

Classify the stated requirement.

Categories: Input / Process / Output

Item	Category
Hours for each item	
Multiply hours by 3	
Add item charges	
Display combined charge	

For one item at 3 per hour for 5 hours, charge is...

2 Break down the solution

Turn right

Key facts in your own words:

Example to trace:

```
HIRE PROGRAM
Read hire details
Calculate charges
Display total
```

Challenge: How could Calculate charges be refined?

Which is a useful sub-problem?

A structure diagram is useful for...

3 Build the running total

Turn around

Key facts in your own words:

Example to trace:

```
total = 0
for item = 1 to 3
    hours = int(input())
    total = total + hours * 3
next item
```

Challenge: What total results from hours 1, 2 and 3?

Order the calculation.

Put these steps in order: Finish the loop / Add hours times 3 to total each time / Repeat input of hours for each item / Set total to 0

Hours 1, 2, 3 at 3 per hour total...

4 Apply a rule once

Turn around

Key facts in your own words:

Example to trace:

```
If total >= 30
    total = total - 5
Then output total
30 becomes 25
```

Challenge: What should happen to totals 29 and 30?

A total of 33 gets 5 off. Final charge?

Total 29 with a discount at >= 30 becomes...

5 Selection inside repetition

Turn left

Key facts in your own words:

Example to trace:

```
For each item:
    if hours > 5 then
        flag long hire
After all items:
    apply order discount
```

Challenge: Where should a per-item long-hire check go?

Which rule belongs after the loop?

A decision inside a loop is...

6 Test the complete design

Turn left

Key facts in your own words:

Example to trace:

Hours: 1, 2, 3 -> 18

Hours: 3, 3, 4 -> 25

Second row total: 30

Discount applied once

Challenge: Why is the second test useful?

Which expected output fits hours 4, 4, 4 with the stated discount?

Hours 3, 3, 4 before the discount total...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Where should `total = 0` be placed?

Hours 2, 2, 5 at 3 per hour. Discount is 5 if `total >= 30`. Output?

Your score

Copy it from the score screen as soon as it appears.

_____ / 20

Key question

How can you turn a long written requirement into a program you can check?

Review and improve

Use review mode. Explain one correction and why the correct answer fits. Implement the full hire program in Python. Test 1, 2, 3 (18), 3, 3, 4 (25) and 4, 4, 4 (31). Compare each output with your design.

Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Identify the inputs, processing and output of the hire program. [3]

b) Write pseudocode for three item-hour inputs at 3 per hour. Subtract 5 if total ≥ 30 , then output the charge. [6]

c) Give inputs that test the exact discount threshold and state the expected output. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Extract requirements from a problem.			
Combine sequence, selection and iteration.			
Refine and test a complete solution.			